

CMOS Inverter & Full Adder

Fundamentals Microelectronics

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This project is prepared for CSCE 3303, section 01



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THE AMERICAN UNIVERSITY IN CAIRO

11/12/2025

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1 Part A: CMOS Inverter

Schematic Overview

In this part, we study the behaviour of a simple CMOS inverter built in a $0.18\mu\text{m}$ and $0.5\mu\text{m}$ technology. The key structural features of the circuit are summarized below:

- The inverter is formed by a complementary pair of transistors: one PMOS device on the top and one NMOS device on the bottom.
- The gates of both transistors are shorted together and driven by the same input signal, which serves as the logic input of the inverter.
- The common drain node of the PMOS and NMOS transistors is the output node of the circuit.
- To represent the load, the output is connected to a series combination of a resistor and a capacitor that both terminate at ground.
- The resistor value is fixed at $1\text{ k}\Omega$, while the capacitor can be either 0.2 pF or 0.5 pF , depending on the CMOS process that is being considered.

A schematic of the inverter together with its load is shown in Figure 1.

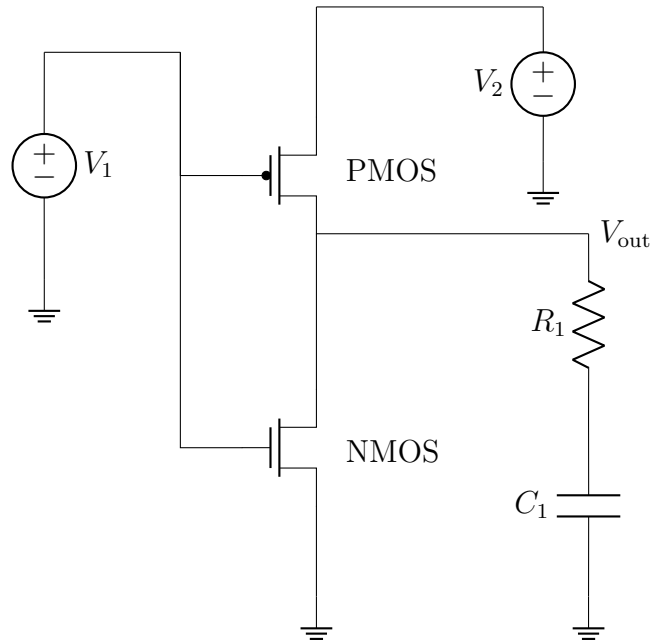


Figure 1: CMOS inverter with PMOS–NMOS stack and series R_1C_1 load at the output node.

1.1 CMOS = 0.18 μm

1.1.1 Matched Design Calculations

To obtain a *matched* inverter, the pull-up (PMOS) and pull-down (NMOS) devices are sized so that they provide comparable drive strength. This is expressed by the condition

$$k'_n \left(\frac{W}{L} \right)_n = k'_p \left(\frac{W}{L} \right)_p, \quad (1)$$

where

- k'_n and k'_p are the process transconductance parameters of the NMOS and PMOS devices, respectively, and
- $(W/L)_n$ and $(W/L)_p$ are the width-to-length ratios of the NMOS and PMOS transistors.

By rearranging this expression, the sizing rule is usually written in terms of the carrier mobilities:

$$\frac{W_p}{W_n} = \frac{\mu_n}{\mu_p}, \quad (2)$$

where

- μ_n is the electron mobility in the NMOS transistor, and
- μ_p is the hole mobility in the PMOS transistor.

This relationship is used to choose the PMOS width so that the inverter has approximately symmetric switching characteristics.

Matched Design Data and Device Sizing

Using the process information and model files provided by the instructor, the NMOS parameters are

$$\begin{aligned} W_n &= 0.9 \mu\text{m}, \\ \mu_n &= 450 \text{ cm}^2/\text{V} \cdot \text{s}, \\ \mu_p &= 100 \text{ cm}^2/\text{V} \cdot \text{s}. \end{aligned}$$

The PMOS width is obtained from the mobility ratio:

$$\begin{aligned} W_p &= W_n \cdot \frac{\mu_n}{\mu_p} \\ W_p &= 0.9 \mu\text{m} \cdot \frac{450}{100} = 4.05 \mu\text{m}. \end{aligned}$$

Source/Drain Area and Perimeter Design Rule. According to the technology documentation, the source/drain extension length X_d is chosen as

$$X_d = 3L_{\min} = 3 \times 0.18 = 0.54 \mu\text{m}.$$

Using this rule, the source/drain areas and perimeters are

- **PMOS:**

$$A_s = A_d = X_d W_p = 0.54 \times 4.05 = 2.187 \mu\text{m}^2,$$

$$P_s = P_d = 2X_d + 2W_p = 2 \times 0.54 + 2 \times 4.05 = 9.18 \mu\text{m}.$$

- **NMOS:**

$$A_s = A_d = X_d W_n = 0.54 \times 0.9 = 0.486 \mu\text{m}^2,$$

$$P_s = P_d = 2X_d + 2W_n = 2 \times 0.54 + 2 \times 0.9 = 2.88 \mu\text{m}.$$

SPICE Netlist for the Matched Inverter

Listing 1: SPICE netlist of the matched CMOS inverter

```
* CMOS inverter, matched design
.include "D:\AUC\Uni\Fall 25\Microelectronics\Project\Saif and adham\CMOS 0.18\MOSFET_models_0p5_0p18-3.inc"
* supply sources
VDD no1 0 DC 1.8V
VG no2 0 DC 0
* MOSFETs
Mpmos no3 no2 no1 no1 PMOSOP18 l=0.18u w=4.05u ad=2.187p as=2.187p pd=9.18u ps=9.18u
Mnmos no3 no2 0 0 NMOSOP18 l=0.18u w=0.9u ad=0.486p as=0.486p pd=2.88u ps=2.88u
* Resistor and Capacitor
Rr1 no3 no4 1k
Cc1 no4 0 0.2p
* DC sweep
.DC VG 0 1.8V 1m
.END
```

VTC Output

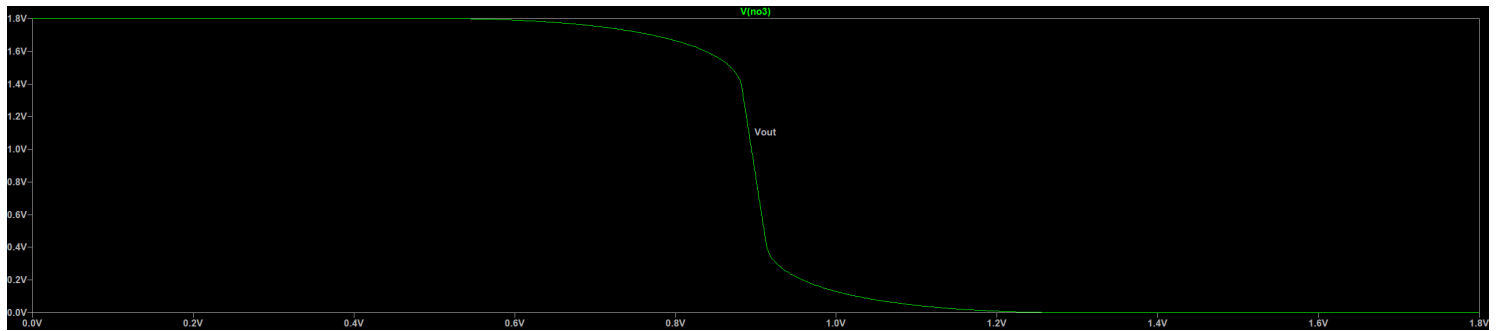


Figure 2: Voltage transfer characteristic (VTC) of the CMOS inverter.

Figure 2 presents the VTC of the CMOS inverter for an input sweep from 0 V to 1.8 V. The behavior can be divided into three main regions:

- **Region 1 – Low input (V_{in} close to 0 V):**
 - **NMOS:** off (cut-off).
 - **PMOS:** strongly on, typically in the triode (linear) region.
 - Result: $V_{out} \approx V_{DD}$ (logic “1”).
- **Region 2 – Transition / switching region:**
 - As V_{in} crosses the NMOS threshold the NMOS turns on and the PMOS begins to turn off; both transistors conduct during this interval.
 - Near the midpoint both devices can operate with substantial channel current (often near saturation), sharing the voltage drop.
 - The VTC exhibits a steep negative slope here (large small-signal gain), producing a sharp switching point.
 - V_{out} therefore falls rapidly from near V_{DD} toward 0 V.
- **Region 3 – High input (V_{in} close to V_{DD}):**
 - **PMOS:** off (cut-off).
 - **NMOS:** strongly on, typically in the triode (linear) region, providing a low-resistance path to ground.
 - Result: $V_{out} \approx 0$ V (logic “0”).

Supply Current Graph

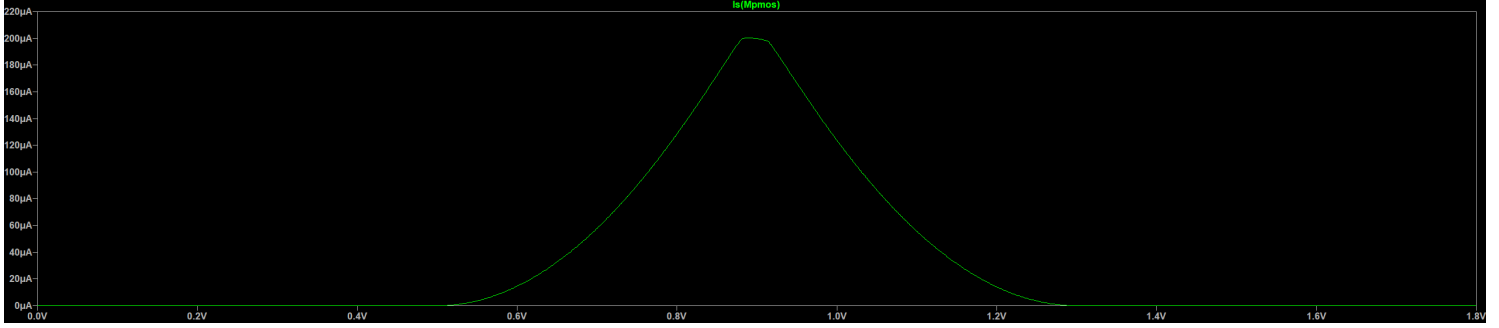


Figure 3: Supply current of the PMOS transistor ($I_S(\text{Mpmos})$) versus input voltage.

Figure 3 shows how the supply current drawn through the PMOS device changes as V_{in} goes from 0 V to 1.8 V. The behavior can be divided into several regions:

- **Region 1 – Low input ($V_{\text{in}} \approx 0$ V):**
 - **NMOS:** cut-off (off).
 - **PMOS:** on, typically in triode since $V_{\text{out}} \approx V_{\text{DD}}$ and therefore $V_{DS,P}$ is small.
 - Supply current is essentially zero, so $I_S(\text{Mpmos}) \approx 0$.
- **Region 2 – Start of the transition ($V_{\text{in}} \approx 0.5$ V):**
 - **NMOS:** begins to turn on and enters saturation.
 - **PMOS:** still conducting, generally in the triode region.
 - A conduction path from V_{DD} through both devices forms, so the supply current starts increasing.
- **Region 3 – Around the switching point ($V_{\text{in}} \approx 0.9$ V):**
 - Both NMOS and PMOS conduct strongly and may both be in saturation for part of this interval.
 - This creates the maximum short-circuit current directly from V_{DD} to ground.
 - Consequently, $I_S(\text{Mpmos})$ reaches its peak value in this region.
- **Region 4 – High input ($V_{\text{in}} \rightarrow 1.8$ V):**
 - **PMOS:** cut-off (off).
 - **NMOS:** in triode, pulling V_{out} close to ground.
 - With the PMOS off, no direct path from V_{DD} to ground remains, so the supply current falls back toward zero.

This confirms that CMOS supply current is negligible in the stable logic states and becomes significant only in the transition region, where both transistors conduct simultaneously.

1.1.2 Minimum Area Design

For the minimum-area design, both NMOS and PMOS are chosen with the smallest allowed width in this technology. We take

$$W_n = W_p = 0.9 \mu\text{m}, \quad L_n = L_p = 0.18 \mu\text{m}.$$

Using the layout rule for the source/drain extension length

$$X_d = 3L_{\min} = 0.54 \mu\text{m},$$

the source/drain area and perimeter are

- **Both devices (minimum area sizing):**

$$A_s = A_d = X_d W = 0.54 \times 0.9 = 0.486 \mu\text{m}^2,$$

$$P_s = P_d = 2X_d + 2W = 2 \times 0.54 + 2 \times 0.9 = 2.88 \mu\text{m}.$$

SPICE Netlist for the Minimum-Area Inverter

Listing 2: SPICE netlist of the minimum-area CMOS inverter

```
* CMOS inverter, minimum-area design
.include "D:\AUC\Uni\Fall 25\Microelectronics\Project\Saif and adham\CMOS 0.18\MOSFET_models_0p5_0p18-3.inc"
* supply sources
VDD no1 0 1.8V
VG no2 0 0
* MOSFETs
Mpmos no3 no2 no1 no1 PMOSOP18 l=0.18u w=0.9u ad=0.486p as=0.486p pd=2.88u ps=2.88u
Mnmos no3 no2 0 0 NMOSOP18 l=0.18u w=0.9u ad=0.486p as=0.486p pd=2.88u ps=2.88u
* Resistor and Capacitor
Rr1 no3 no4 1k
Cc1 no4 0 0.2p
* DC sweep
.DC VG 0 1.8V 1m
.END
```


VTC Output

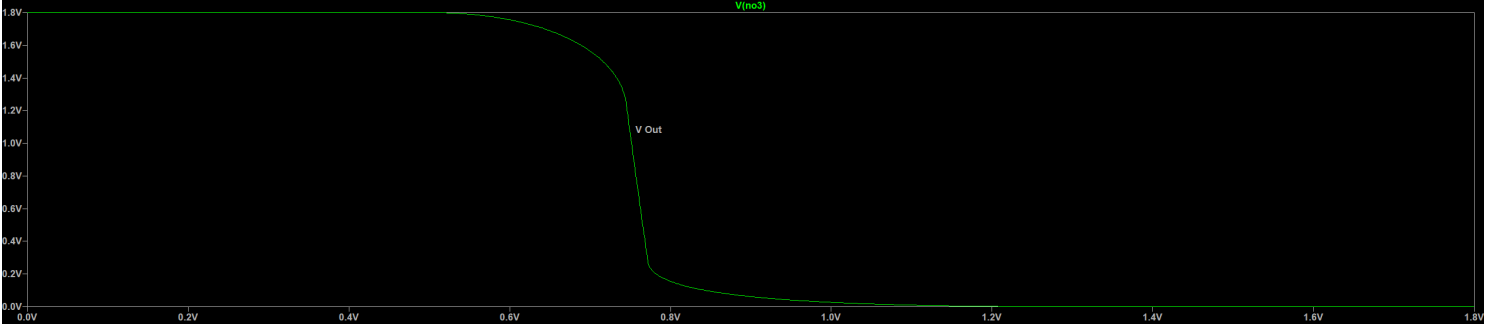


Figure 4: Voltage transfer characteristic (VTC) of the minimum-area CMOS inverter.

Figure 4 shows the VTC of the minimum-area inverter, where both NMOS and PMOS use the same minimum width $W = 0.9 \mu\text{m}$. The behavior can be summarized as follows:

- **Switching threshold:**
 - Compared to the matched-width design, the switching point moves to a lower V_{in} .
 - This happens because the NMOS and PMOS now have equal width but different mobilities; the NMOS is effectively stronger, so the output flips earlier as V_{in} increases.
- **Shape of the transition:**
 - The output still transitions from a high level near V_{DD} to a low level near 0 V.
 - However, the transition is less symmetric than in the matched case and the steep region is biased toward lower V_{in} .
- **Logic levels and noise margins:**
 - The high and low output levels remain clearly defined, so the circuit still behaves as a valid inverter.
 - In the matched design, the sizing is chosen such that

$$k'_n \left(\frac{W}{L} \right)_n \approx k'_p \left(\frac{W}{L} \right)_p,$$

which makes the current balance

$$I_{Dn} \approx \frac{1}{2} k'_n \left(\frac{W}{L} \right)_n (V_M - V_{Tn})^2 \approx \frac{1}{2} k'_p \left(\frac{W}{L} \right)_p (V_{DD} - V_M - |V_{Tp}|)^2$$

roughly symmetric and gives a switching voltage $V_M \approx V_{DD}/2$, so the noise margins NM_H and NM_L are relatively large.

- In the minimum-area design, we set $W_n = W_p$ (and $L_n = L_p$), so the effective strengths satisfy

$$k'_n \left(\frac{W}{L} \right)_n \gg k'_p \left(\frac{W}{L} \right)_p$$

because $k'_n \gg k'_p$ (reflecting $\mu_n \gg \mu_p$). This makes the NMOS stronger, pulls the switching point down ($V_M < V_{DD}/2$), and shifts V_{IL} and V_{IH} toward lower values of V_{in} .

- Since $V_{OH} \approx V_{DD}$ and $V_{OL} \approx 0$ stay almost unchanged, the high-level noise margin $NM_H = V_{OH} - V_{IH}$ becomes smaller (while $NM_L = V_{IL} - V_{OL}$ may increase slightly), so the minimum-area inverter has an overall reduced noise margin compared to the matched design.

Supply Current Graph

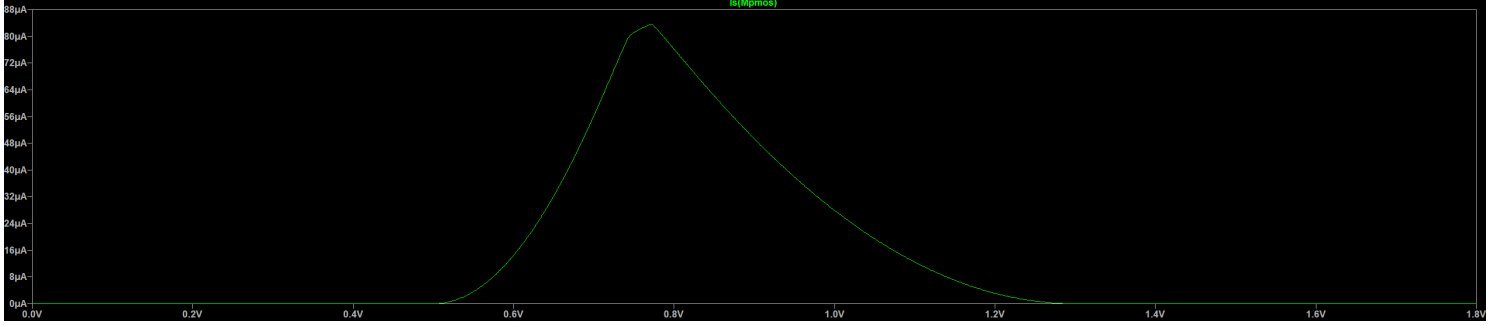


Figure 5: Supply current of the PMOS transistor ($I_S(\text{Mpmos})$) for the minimum-area inverter.

Figure 5 plots the supply current for the minimum-area configuration. Its main features are:

- **Current in the stable regions:**
 - For low V_{in} and for high V_{in} , one of the transistors is off, so the direct path from V_{DD} to ground is cut.
 - As a result, the supply current is close to zero in both logic states.
- **Peak around the transition:**
 - A single current peak appears in the switching region, where both NMOS and PMOS conduct at the same time.
 - The peak is located near the lower switching threshold of the minimum-area inverter, so it is slightly shifted compared to the matched design.
- **Effect of minimum sizing:**
 - Because the devices are narrower, it reduces the magnitude of the peak supply current compared to the matched-width case.

1.2 Propagation Delay

1.2.1 Matched Design

Netlist

Listing 3: LTspice netlist used for propagation delay of the matched CMOS inverter

```
* CMOS inverter transient analysis, matched design
.include "D:\AUC\Uni\Fall 25\Microelectronics\Project\Saif and adham\CMOS 0.18\MOSFET_models_0p5_0p18-3.inc"
* supply sources
VDD no1 0 1.8V
VG no2 0 pulse(0v 1.8v 0 1p 1p 5n 10n)
* MOSFETs
Mpmos no3 no2 no1 no1 PMOSOP18 l=0.18u w=4.05u ad=2.187p as=2.187p pd=9.18u ps=9.18u
Mnmos no3 no2 0 0 NMOSOP18 l=0.18u w=0.9u ad=0.486p as=0.486p pd=2.88u ps=2.88u
* Resistor and Capacitor
Rr1 no3 no4 1k
Cc1 no4 0 0.2p
* Transient analysis
.tran 1p 10n
.measure tran TphL trig V(no2) val=0.9 rise=1 targ V(no4) val=0.9 fall=1
.measure tran TplH trig V(no2) val=0.9 fall=1 targ V(no4) val=0.9 rise=1
.measure Tp param= '(TphL+TplH)/2'
.END
```

Propagation Delay Waveform

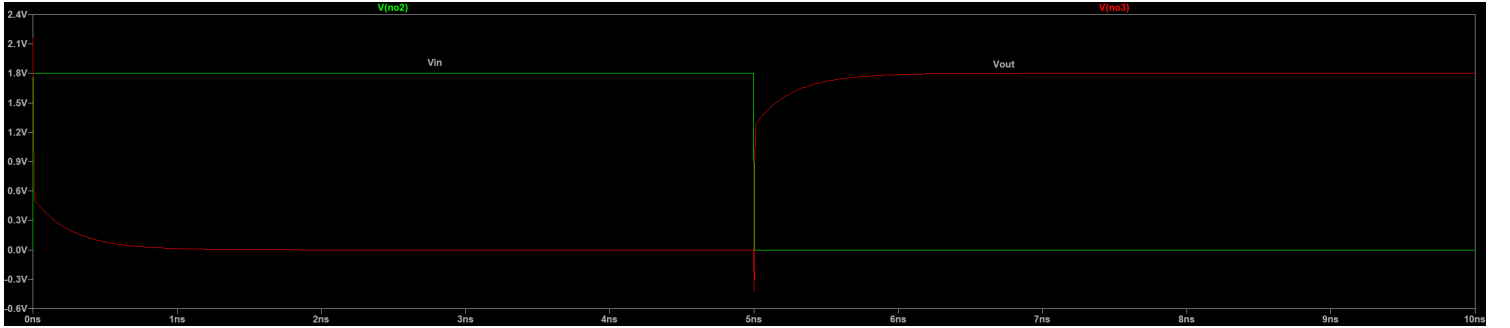


Figure 6: Transient response of the matched 0.18 μm CMOS inverter used to extract the propagation delay.

Delay Definitions and Analytical Background

For a CMOS inverter, two propagation delays are defined at the 50% voltage level (0.9 V for $V_{DD} = 1.8$ V):

t_{pHL} : input makes a low-to-high transition, output falls,

t_{pLH} : input makes a high-to-low transition, output rises.

In LTspice, these are extracted by equating the time instants where the input and output waveforms cross 0.9 V:

$$\begin{aligned}
t_{pHL} : \quad & V_{in}(t_{TRIG}) = 0.9 \text{ V (first rise),} \\
& V_{out}(t_{TARG}) = 0.9 \text{ V (first fall),} \\
& t_{pHL} = t_{TARG} - t_{TRIG}; \\
t_{pLH} : \quad & V_{in}(t_{TRIG}) = 0.9 \text{ V (first fall),} \\
& V_{out}(t_{TARG}) = 0.9 \text{ V (first rise),} \\
& t_{pLH} = t_{TARG} - t_{TRIG}.
\end{aligned}$$

Simulation Results and Interpretation

Using the `.measure` statements in Listing 3, LTspice reports the following results in the SPICE Output Log:

```

tphl=1.97718603732e-10 FROM 4.99999993017e-13 TO 1.98218603725e-10
tplh=1.98138216386e-10 FROM 5.00149999999e-09 TO 5.19963821637e-09
tp: ' (TpHL+TpLH) /2 '=1.97928410059e-10

```

Figure 7: SPICE Output Log.

- High-to-low delay:

$$t_{pHL} = 197.72 \text{ ps} \quad (4.99999993017 \times 10^{-13} \text{ s} \rightarrow 1.98218603725 \times 10^{-10} \text{ s}),$$

- Low-to-high delay:

$$t_{pLH} = 198.14 \text{ ps} \quad (5.00149999999 \times 10^{-9} \text{ s} \rightarrow 5.19963821637 \times 10^{-9} \text{ s}),$$

- Average propagation delay:

$$t_p = \frac{t_{pHL} + t_{pLH}}{2} = 197.93 \text{ ps}.$$

The fact that t_{pHL} and t_{pLH} are almost identical (difference < 1 ps) confirms that the matched sizing produces a nearly symmetric inverter: the effective pull-up and pull-down strengths are balanced, and the inverter exhibits an average propagation delay of approximately 198 ps under the specified loading conditions.

1.2.2 Minimum Area Design Propagation Delay

Netlist

Listing 4: LTspice netlist for propagation delay of the minimum-area CMOS inverter

```
* CMOS inverter transient analysis, minimum area design
.include "D:\AUC\Uni\Fall 25\Microelectronics\Project\Saif and adham\CMOS 0.18\MOSFET_models_0p5_0p18-3.inc"
* supply sources
VDD no1 0 1.8V
VG no2 0 pulse(0v 1.8v 0 1p 1p 5n 10n)
* MOSFETs
Mpmos no3 no2 no1 no1 PMOSOP18 l=0.18u w=0.9u ad=0.486p as=0.486p pd=2.88u ps=2.88u
Mnmos no3 no2 0 0 NMOSOP18 l=0.18u w=0.9u ad=0.486p as=0.486p pd=2.88u ps=2.88u
* Resistor and Capacitor
Rr1 no3 no4 1k
Cc1 no4 0 0.2p
* Transient analysis
.tran 1p 10n
.measure tran TpHL trig V(no2) val=0.9 rise=1 targ V(no4) val=0.9 fall=1
.measure tran TpLH trig V(no2) val=0.9 fall=1 targ V(no4) val=0.9 rise=1
.measure Tp param= '(TpHL+TpLH)/2'
.END
```

Propagation Delay Waveform

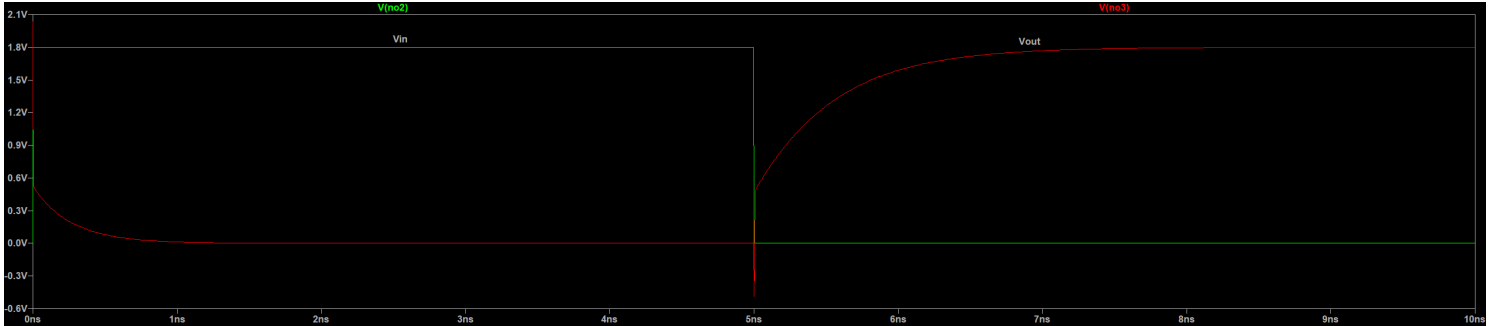


Figure 8: Transient response of the minimum-area 0.18 μm CMOS inverter used to extract the propagation delay.

Propagation Delay Results

Using the `.measure` statements in Listing 3, LTspice reports the following results in the SPICE Output Log:

```

tphl=1.94953879139e-10 FROM 5.00000013694e-13 TO 1.95453879152e-10
tplh=4.48002495875e-10 FROM 5.00149999999e-09 TO 5.44950249586e-09
tp: '(Tphl+Tplh)/2'=3.21478187507e-10

```

Figure 9: SPICE Output Log.

- High-to-low delay:

$$t_{pHL} = 194.95 \text{ ps} \quad (5.00000013694 \times 10^{-13} \text{ s} \rightarrow 1.95453879152 \times 10^{-10} \text{ s}),$$

- Low-to-high delay:

$$t_{pLH} = 448.00 \text{ ps} \quad (5.00149999999 \times 10^{-9} \text{ s} \rightarrow 5.44950249586 \times 10^{-9} \text{ s}),$$

- Average propagation delay:

$$t_p = \frac{t_{pHL} + t_{pLH}}{2} = 321.48 \text{ ps}.$$

$$t_{pHL} \ll t_{pLH}, \quad t_p \text{ is larger than in the matched case.}$$

Analysis:

- The minimum-area inverter reduces device size, but this comes at the cost of slower signal transitions.
- A large imbalance appears between the high-to-low and low-to-high delays.
- The much larger t_{pLH} is caused by the weak PMOS drive strength at minimum width.
- Since PMOS mobility is lower than NMOS, the low-to-high transition becomes noticeably slower.
- Overall, the reduced sizing improves area but increases delay, giving an average propagation delay of about 321.5 ps.

1.3 CMOS = 0.5 μm

1.3.1 Matched Design Calculations

For the matched design in 0.5 μm CMOS technology, the NMOS width is given as $W_n = 1.25 \mu\text{m}$. The PMOS width is again chosen to match the drive strength of the NMOS using the mobility-based sizing rule

$$\frac{W_p}{W_n} = \frac{\mu_n}{\mu_p}, \tag{3}$$

where μ_n and μ_p are the electron and hole mobilities in the 0.5 μm process, respectively.

Taking the new model parameters

$$\mu_n = 460 \text{ cm}^2/\text{V} \cdot \text{s}, \quad \mu_p = 115 \text{ cm}^2/\text{V} \cdot \text{s},$$

the PMOS width becomes

$$W_p = W_n \cdot \frac{\mu_n}{\mu_p}$$
$$W_p = 1.25 \text{ } \mu\text{m} \cdot \frac{460}{115} = 5 \text{ } \mu\text{m}.$$

Source/Drain Area and Perimeter Design Rule. According to the layout design rule, the source/drain extension length is chosen as

$$X_d = 3L_{\min} = 3 \times 0.5 = 1.5 \text{ } \mu\text{m}.$$

the source/drain area and perimeter for each device are

- **PMOS:**

$$A_s = A_d = XW_p = 1.5 \times 5 = 7.5 \text{ } \mu\text{m}^2,$$

$$P_s = P_d = 2X + 2W_p = 2 \times 1.5 + 2 \times 5 = 13 \text{ } \mu\text{m}.$$

- **NMOS:**

$$A_s = A_d = X_dW_n = 1.5 \times 1.25 = 1.875 \text{ } \mu\text{m}^2,$$

$$P_s = P_d = 2X_d + 2W_n = 2 \times 1.5 + 2 \times 1.25 = 5.5 \text{ } \mu\text{m}.$$

SPICE Netlist for the Matched 0.5 μm Inverter

Listing 5: SPICE netlist of the matched 0.5 μm CMOS inverter

```
* CMOS inverter, matched design 0.5um
.include "D:\AUC\Uni\Fall 25\Microelectronics\Project\Saif and adham\CMOS 0.18\MOSFET_models_0p5_0p18-3.inc"
* supply sources
VDD no1 0 3.3V
VG no2 0 0
* MOSFETs
Mpmos no3 no2 no1 no1 PMOSOP5 l=0.5u w=5u ad=7.5p as=7.5p pd=13u ps=13u
Mnmos no3 no2 0 0 NMOSOP5 l=0.5u w=1.25u ad=1.875p as=1.875p pd=5.5u ps=5.5u
* Resistor and Capacitor
Rr1 no3 no4 1k
Cc1 no4 0 0.5p
.DC VG 0 3.3V 1m
.END
```

VTC Output (Matched 0.5 μm Design)

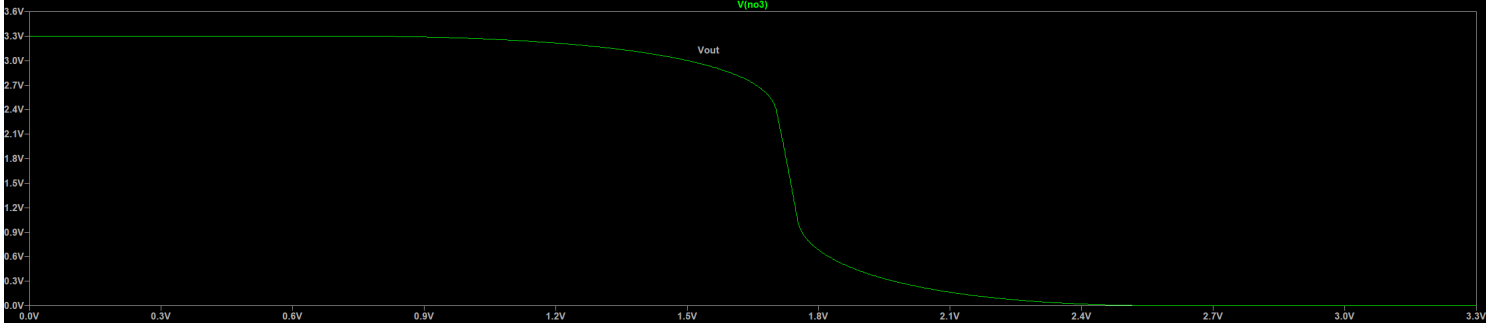


Figure 10: Voltage transfer characteristic (VTC) of the matched 0.5 μm CMOS inverter.

Figure 10 shows the VTC of the matched 0.5 μm CMOS inverter with $V_{DD} = 3.3\text{ V}$. Qualitatively, the curve has the same three regions discussed for the 0.18 μm inverter (high output at low V_{in} , steep transition region, and low output at high V_{in}). The main points specific to this technology node are:

- The device widths are chosen using the mobility-based matching rule $\frac{W_p}{W_n} = \frac{\mu_n}{\mu_p}$, which gives $W_n = 1.25\text{ }\mu\text{m}$ and $W_p \approx 5\text{ }\mu\text{m}$. This balances the effective strengths of the pull-up and pull-down networks.
- As a result, the switching threshold V_M appears close to mid-supply, $V_{DD}/2 \approx 1.65\text{ V}$, similar to the matched 0.18 μm design. The transition region is steep and centered, which yields good noise margins.
- The output still swings very close to the rails ($V_{OH} \approx 3.3\text{ V}$, $V_{OL} \approx 0\text{ V}$), but the larger supply voltage and wider devices lead to a somewhat sharper transition compared with the 0.18 μm case.

Overall, the matched 0.5 μm inverter preserves the symmetric switching behavior of the 0.18 μm matched design, but with a higher absolute switching voltage due to the larger V_{DD} .

Supply Current Graph (Matched 0.5 μm Design)

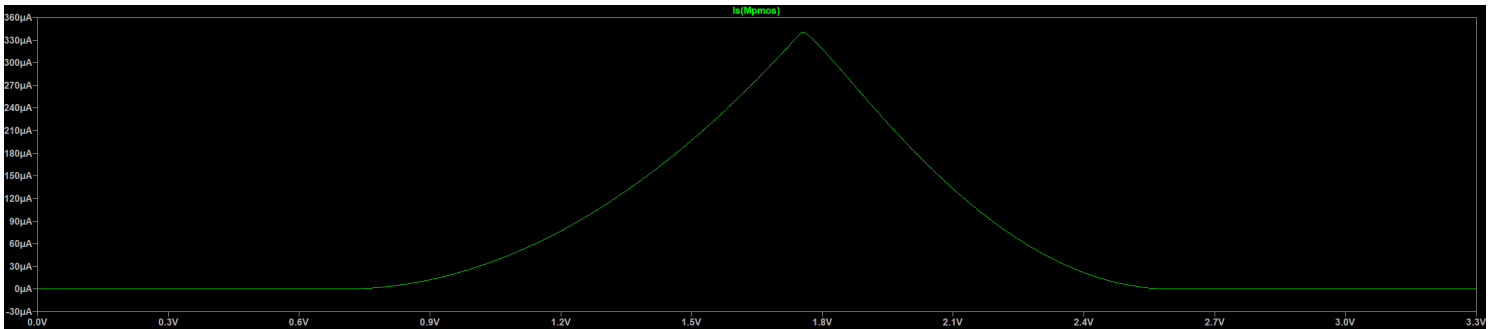


Figure 11: Supply current of the PMOS transistor ($I_S(\text{Mpmos})$) versus input voltage.

Figure 11 plots the simulated supply current of the PMOS device, $I_S(\text{Mpmos})$, as V_{in} is swept from 0 to 3.3 V. The qualitative behavior is the same as in the 0.18 μm matched case:

- At low and high input levels, one of the transistors is off, so there is no direct path from V_{DD} to ground and the supply current is nearly zero in both stable logic states.
- A single pronounced peak appears in the transition region, where both NMOS and PMOS conduct simultaneously and a short-circuit path exists between the supply and ground.
- Compared with the $0.18\text{ }\mu\text{m}$ design, the peak current is higher. This is consistent with the larger supply voltage (3.3 V instead of 1.8 V) and the wider devices, which together increase the available drive strength.

Thus, the matched $0.5\text{ }\mu\text{m}$ inverter shows the same current profile as the $0.18\text{ }\mu\text{m}$ matched design, but with a larger current peak due to the higher V_{DD} and transistor sizes.

1.3.2 Minimum Area Design for CMOS = $0.5\text{ }\mu\text{m}$

For the minimum-area $0.5\text{ }\mu\text{m}$ design, both NMOS and PMOS devices are chosen with the smallest allowed geometry in this technology:

$$W_n = W_p = 1.25\text{ }\mu\text{m}, \quad L_n = L_p = 0.5\text{ }\mu\text{m}.$$

Source/Drain Area and Perimeter. Using the layout rule

$$X_d = 3L_{\min} = 3 \times 0.5 = 1.5\text{ }\mu\text{m}.$$

the common source/drain area and perimeter for both transistors become

- **Both devices (minimum area sizing):**

$$A_s = A_d = X_d W = 1.5 \times 1.25 = 1.875\text{ }\mu\text{m}^2,$$

$$P_s = P_d = 2X_d + 2W = 2 \times 1.5 + 2 \times 1.25 = 5.5\text{ }\mu\text{m}.$$

SPICE Netlist for the Minimum-Area $0.5\text{ }\mu\text{m}$ Inverter

Listing 6: SPICE netlist of the minimum-area $0.5\text{ }\mu\text{m}$ CMOS inverter

```
* CMOS inverter, minimum area 0.5um
.include "D:\AUC\Uni\Fall 25\Microelectronics\Project\Saif and adham\CMOS 0.18\MOSFET_models_0p5_0p18-3.inc"
* supply sources
VDD no1 0 3.3V
VG no2 0 0
* MOSFETs
Mpmos no3 no2 no1 no1 PMOS0P5 l=0.5u w=1.25u ad=1.875p as=1.875p pd=5.5u ps=5.5u
Mnmos no3 no2 0 0 NMOS0P5 l=0.5u w=1.25u ad=1.875p as=1.875p pd=5.5u ps=5.5u
* Resistor and Capacitor
```

```

Rr1 no3 no4 1k
Cc1 no4 0 0.5p
.DC VG 0 3.3V 1m
.END

```

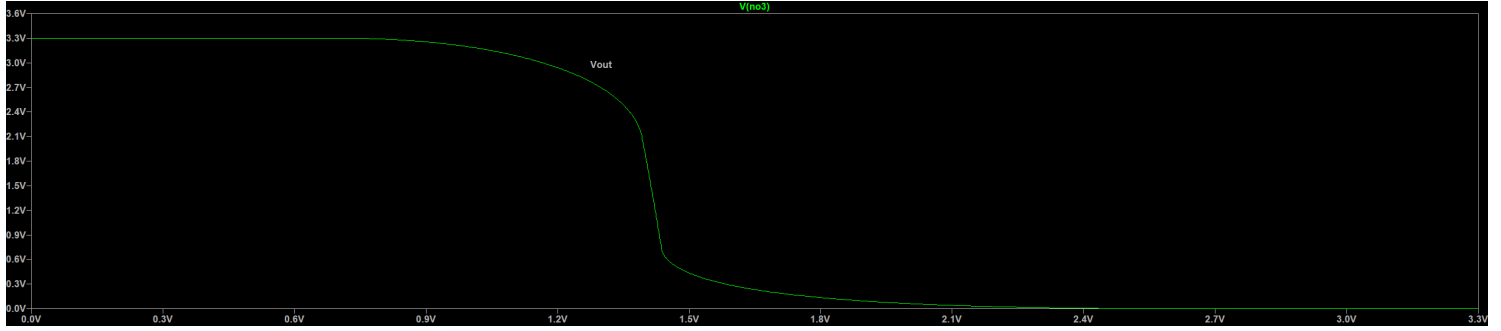


Figure 12: Voltage transfer characteristic (VTC) of the minimum-area $0.5\ \mu\text{m}$ CMOS inverter.

Figure 12 shows the VTC of the minimum-area $0.5\ \mu\text{m}$ inverter. Qualitatively, the curve has the same three regions as in the $0.18\ \mu\text{m}$ case (high output at low V_{in} , a steep transition region, and low output at high V_{in}), but with some important differences:

- Because both devices use the minimum width $W_n = W_p = 1.25\ \mu\text{m}$ while $\mu_n > \mu_p$, the NMOS is stronger than the PMOS. This shifts the switching threshold V_M to a value lower than $V_{DD}/2$ (for $V_{DD} = 3.3\ \text{V}$), so the output flips earlier as V_{in} increases.
- The transition is therefore less symmetric around mid-supply compared with the matched-width design, which was sized using $\frac{W_p}{W_n} = \frac{\mu_n}{\mu_p}$.
- High and low output levels still settle close to $V_{OH} \approx 3.3\ \text{V}$ and $V_{OL} \approx 0\ \text{V}$, but the shift of V_M , V_{IL} and V_{IH} toward lower input voltages reduces the high-level noise margin $\text{NM}_H = V_{OH} - V_{IH}$, while $\text{NM}_L = V_{IL} - V_{OL}$ may increase slightly.

Overall, the minimum-area $0.5\ \mu\text{m}$ inverter behaves like a normal inverter but with a stronger pull-down, a lower switching point and slightly worse noise margins than the matched design.

Supply Current Graph

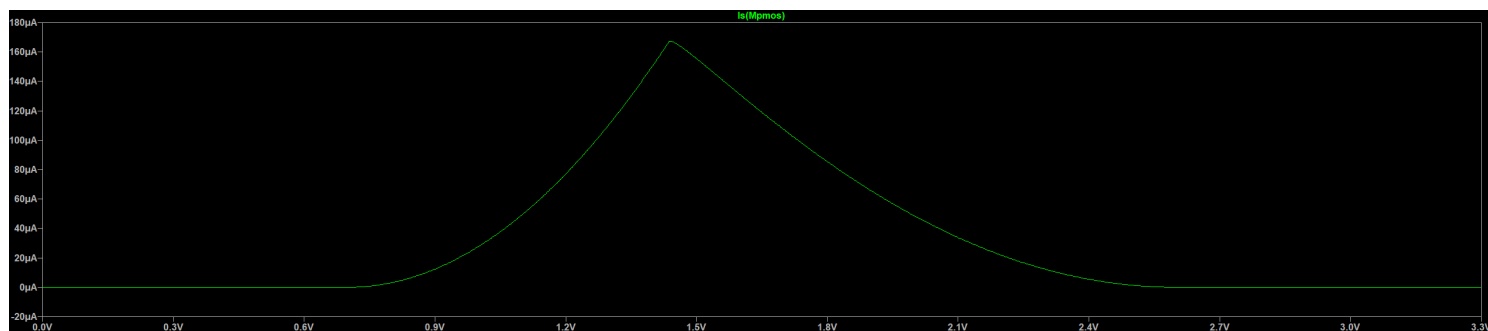


Figure 13: Supply current of the PMOS transistor ($I_S(\text{Mpmos})$) for the minimum-area $0.5 \mu\text{m}$ inverter.

Figure 13 plots the supply current of the PMOS device $I_S(\text{Mpmos})$ versus V_{in} for the minimum-area $0.5 \mu\text{m}$ design. The qualitative shape is the same as in the $0.18 \mu\text{m}$ case:

- For low and high input levels, one of the transistors is off, so there is no direct path from V_{DD} to ground and the supply current is almost zero.
- In the transition region, both devices conduct simultaneously, creating a short-circuit path and producing a single current peak around the switching point.
- Because the PMOS width here is minimum (compared with the wider matched PMOS), its drive strength is reduced, so the peak current is smaller than in the matched $0.5 \mu\text{m}$ inverter.

Thus, minimum-area sizing at $0.5 \mu\text{m}$ keeps the same basic behavior as the $0.18 \mu\text{m}$ inverter but with a shifted VTC, lower peak supply current and tighter noise margins.

1.3.3 Matched Propagation Delay (CMOS = $0.5 \mu\text{m}$)

Netlist

Listing 7: LTspice netlist used for propagation delay of the matched $0.5 \mu\text{m}$ CMOS inverter

```
* CMOS inverter transient analysis, matched design 0.5um
.include "D:\AUC\Uni\Fall 25\Microelectronics\Project\Saif and adham\CMOS 0.18\MOSFET_models_0p5_0p18-3.inc"
* supply sources
VDD no1 0 3.3V
VG no2 0 pulse(0v 3.3v 0 1p 1p 5n 10n)
* MOSFETs
Mpmos no3 no2 no1 no1 PMOSOP5 l=0.5u w=5u ad=7.5p as=7.5p pd=13u ps=13u
Mnmos no3 no2 0 0 NMOSOP5 l=0.5u w=1.25u ad=1.875p as=1.875p pd=5.5u ps=5.5u
* Resistor and Capacitor
Rr1 no3 no4 1k
```

```

Cc1 no4 0 0.5p
* Transient analysis
.tran 1p 10n
.measure tran TphL trig V(no2) val=1.65 rise=1 targ V(no4) val=1.65 fall=1
.measure tran TplH trig V(no2) val=1.65 fall=1 targ V(no4) val=1.65 rise=1
.measure Tp param= '(TphL+TplH)/2'
.END

```

Propagation Delay Waveform

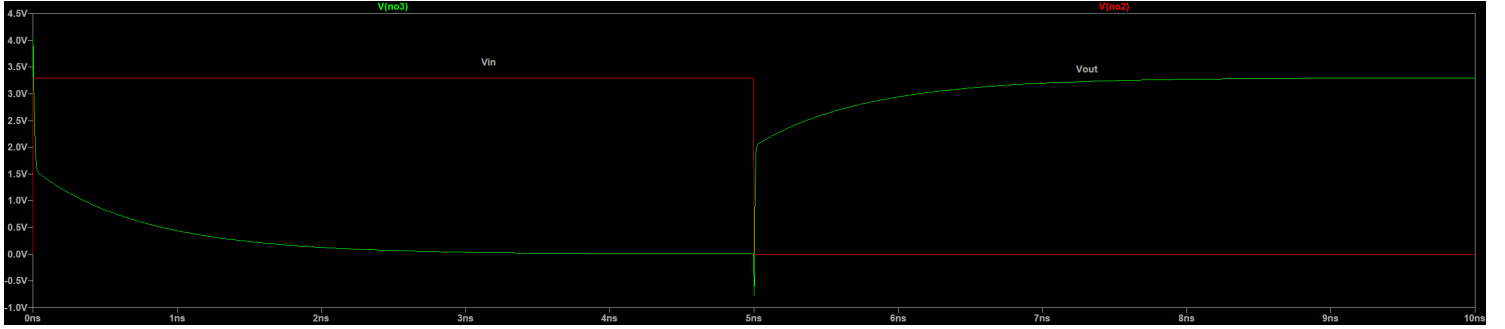


Figure 14: Transient response of the matched 0.5 μm CMOS inverter used to extract the propagation delay.

For a CMOS inverter, the propagation delays are defined at the 50% voltage level. In this technology, $V_{DD} = 3.3\text{ V}$, so the mid-level is

$$V_M = \frac{V_{DD}}{2} = 1.65\text{ V}.$$

Simulation Results and Interpretation

Using the `.measure` statements in Listing 7, LTspice reports the following results in the SPICE Output Log:

```

tphl=6.36489731719e-10 FROM 5.00000015232e-13 TO 6.36989731734e-10
tplh=5.63234362481e-10 FROM 5.00149999999e-09 TO 5.56473436247e-09
tp: '(TphL+TplH)/2'=5.998620471e-10

```

Figure 15: SPICE Output Log.

- High-to-low delay:

$$t_{pHL} = 636.49\text{ ps} \quad (5.00000015232 \times 10^{-13}\text{ s} \rightarrow 6.36989731734 \times 10^{-10}\text{ s}),$$

- Low-to-high delay:

$$t_{pLH} = 563.23\text{ ps} \quad (5.00149999999 \times 10^{-9}\text{ s} \rightarrow 5.56473436247 \times 10^{-9}\text{ s}),$$

- Average propagation delay:

$$t_p = \frac{t_{pHL} + t_{pLH}}{2} = 599.86 \text{ ps.}$$

Compared with the 0.18 μm case, the propagation delays are significantly larger (599.86 ps instead of ≈ 197.93 ps), mainly due to the larger technology and the higher load capacitance (0.5 pF instead of 0.2 pF).

1.3.4 Minimum-Area Design Propagation Delay

Netlist

Listing 8: LTspice netlist for propagation delay of the minimum-area 0.5 μm CMOS inverter

```
* CMOS inverter transient analysis, minimum area 0.5um
.include "D:\AUC\Uni\Fall 25\Microelectronics\Project\Saif and adham\CMOS 0.18\MOSFET_models_0p5_0p18-3.inc"
* supply sources
VDD no1 0 3.3V
VG no2 0 pulse(0v 3.3v 0 1p 1p 5n 10n)
* MOSFETs
Mpmos no3 no2 no1 no1 PMOSOP5 l=0.5u w=1.25u ad=1.875p as=1.875p pd=5.5u ps=5.5u
Mnmos no3 no2 0 0 NMOSOP5 l=0.5u w=1.25u ad=1.875p as=1.875p pd=5.5u ps=5.5u
* Resistor and Capacitor
Rr1 no3 no4 1k
Cc1 no4 0 0.5p
* Transient analysis
.tran 1p 10n
.measure tran TpHL trig V(no2) val=1.65 rise=1 targ V(no4) val=1.65 fall=1
.measure tran TpLH trig V(no2) val=1.65 fall=1 targ V(no4) val=1.65 rise=1
.measure Tp param= '(TpHL+TpLH)/2'
.END
```

Propagation Delay Waveform

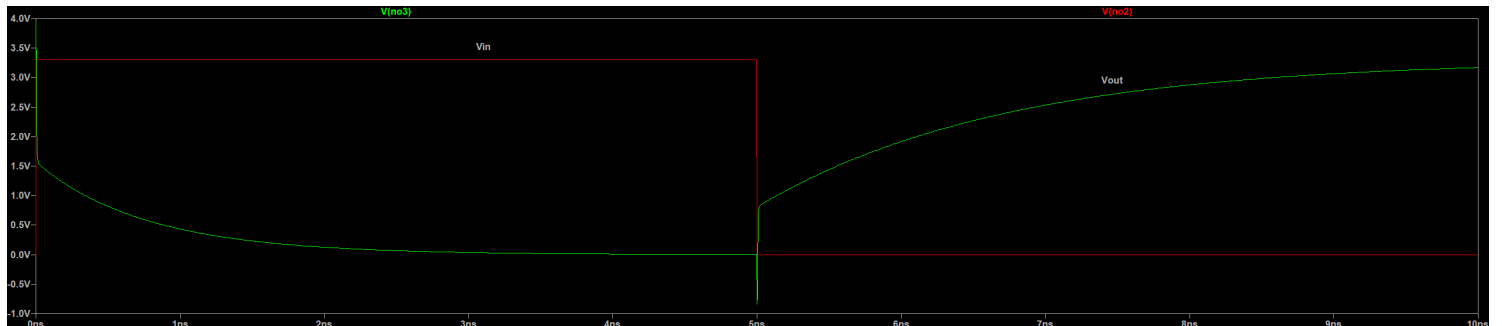


Figure 16: Transient response of the minimum-area 0.5 μm CMOS inverter used to extract the propagation delay.

Propagation Delay Results and Equivalent Resistances

Using the `.measure` statements in Listing 8, LTspice reports the following results in the SPICE Output Log:

```
tphl=6.26283346999e-10 FROM 4.99999987792e-13 TO 6.26783346987e-10
tplh=1.27410247716e-09 FROM 5.00149999999e-09 TO 6.27560247715e-09
tp: ' (TpHL+TpLH) /2 '=9.5019291208e-10
```

Figure 17: SPICE Output Log.

- High-to-low delay:

$$t_{pHL} = 626.28 \text{ ps} \quad (4.99999987792 \times 10^{-13} \text{ s} \rightarrow 6.26783346987 \times 10^{-10} \text{ s}),$$

- Low-to-high delay:

$$t_{pLH} = 1274.10 \text{ ps} \quad (5.00149999999 \times 10^{-9} \text{ s} \rightarrow 6.27560247715 \times 10^{-9} \text{ s}),$$

- Average propagation delay:

$$t_p = \frac{t_{pHL} + t_{pLH}}{2} = 950.19 \text{ ps}.$$

$$t_{pHL} \ll t_{pLH}, \quad t_p \text{ is significantly larger than in the matched case,}$$

- At the $0.5 \mu\text{m}$ node, using minimum-size devices saves area but slows the rising transition and increases overall propagation delay.
- The delays become highly asymmetric, with a much larger t_{pLH} than t_{pHL} .
- The slow low-to-high transition is due to the weaker PMOS drive strength, since both NMOS and PMOS use minimum widths and PMOS mobility is inherently lower.
- As a result, the output takes significantly longer to rise than to fall.

2 Part B: Full Adder

Schematic Design

The full adder is a combinational digital circuit with three binary inputs (A, B, C) and two outputs: the Sum (S) and the Carry-out (C_{out}). The logic functions are

$$S = AB'C' + A'BC' + A'B'C + ABC,$$

$$C_{\text{out}} = AB + AC + BC.$$

Here, A and B are the primary operand bits and C represents the carry-in from a less significant stage, while C_{out} is the carry propagated to the next stage of addition.

PUN is obtained by finding S in terms of complemented inputs
 PDN is obtained by finding $\bar{S}$ in terms of uncomplemented inputs

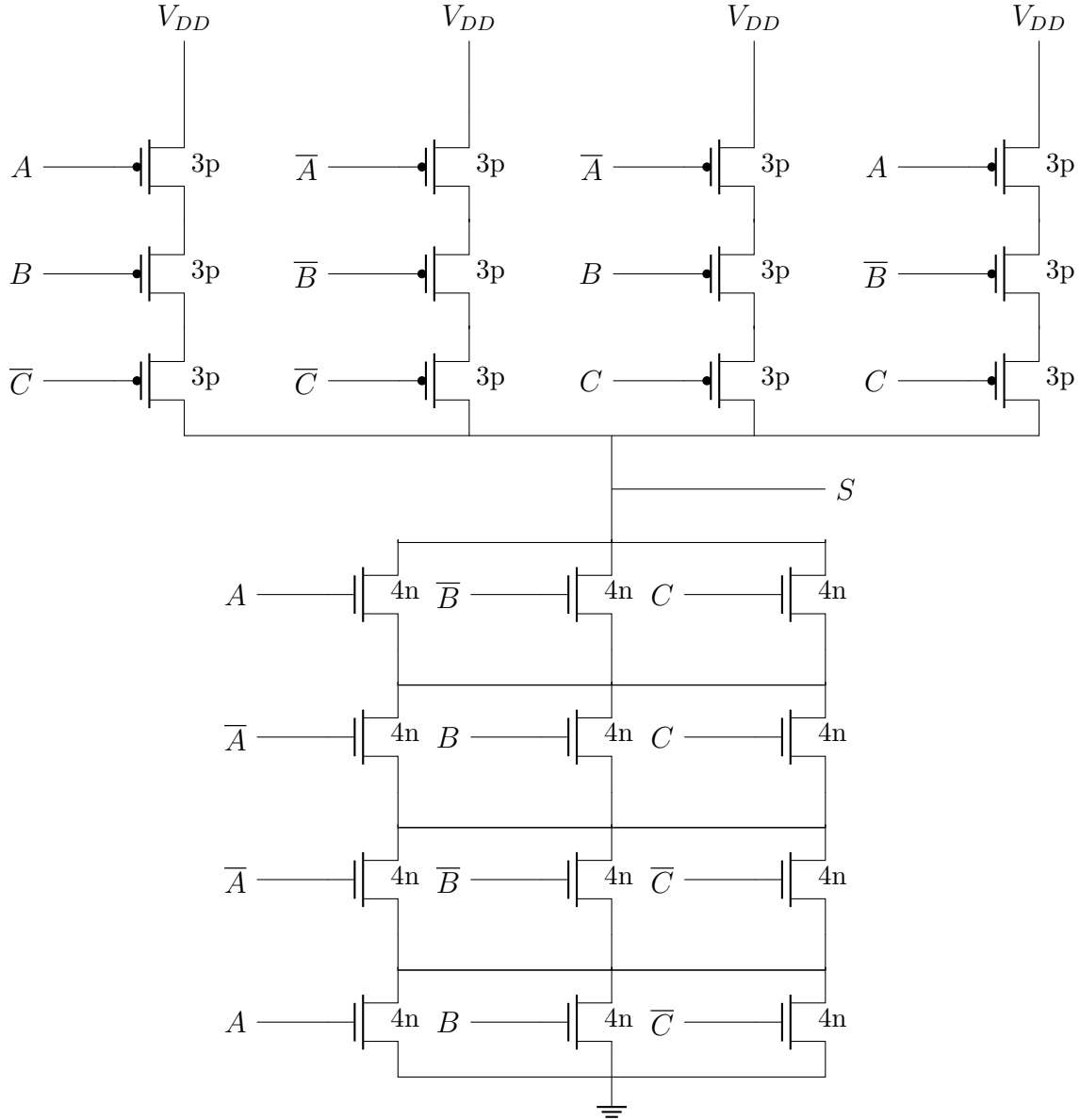


Figure 18: CMOS full-adder Sum output network (PUN and PDN), 36 Transistors

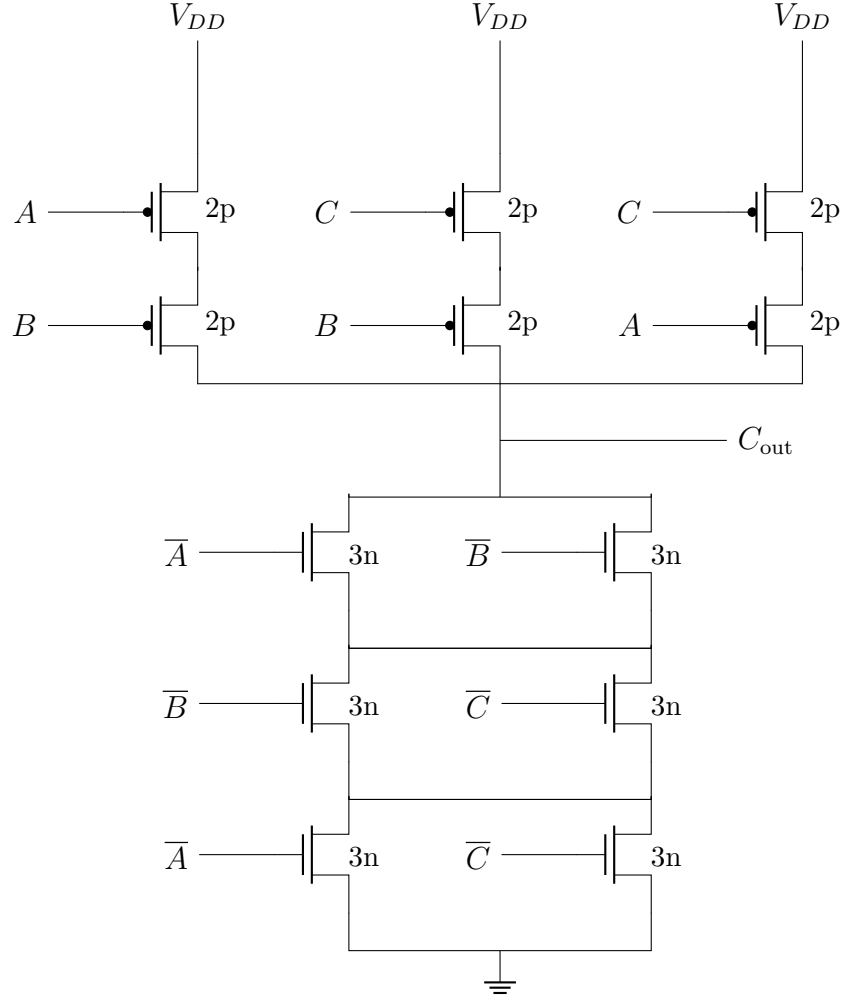


Figure 19: CMOS full-adder Carry-out output network (PUN and PDN), 18 transistors

2.1 Device Dimensions

2.1.1 Sizing for $0.18 \mu\text{m}$ Transistor

Throughout the design we work with minimum area inverters as we did in **Part A**. Their dimensions are

$$W_n = 0.9 \mu\text{m}, \quad W_p = 0.9 \mu\text{m},$$

$$L_n = L_p = 0.18 \mu\text{m}, \quad X_d = 3L_{\min} = 3 \times 0.18 = 0.54 \mu\text{m}.$$

The corresponding diffusion regions (source and drain are taken equal) for these minimum area inverters are

$$A_s = A_d = X_d W_n = 0.54 \times 0.9 = 0.486 \mu\text{m}^2,$$

$$P_s = P_d = 2X_d + 2W_n = 2 \times 0.54 + 2 \times 0.9 = 2.88 \mu\text{m}.$$

These area and perimeter values are reused for all transistors that implement the simple inversions of A , B and C .

2.1.2 Sizing for the Full-Adder Output Transistors

For Part B, the full-adder output is intentionally upsized with respect to the minimum inverter. The specification requires an effective aspect ratio $n = 5$ for NMOS devices and $p = 2n = 10$ for PMOS devices, with the channel length kept at the $0.18 \mu\text{m}$ minimum:

$$\left(\frac{W}{L}\right)_n = n = 5, \quad L_n = 0.18 \mu\text{m}, \quad W_n = nL_n = 5 \times 0.18 = 0.9 \mu\text{m},$$
$$\left(\frac{W}{L}\right)_p = p = 10, \quad L_p = 0.18 \mu\text{m}, \quad W_p = pL_p = 10 \times 0.18 = 1.8 \mu\text{m}.$$

Hence every NMOS in the Sum output branch uses $W_n = 0.9 \mu\text{m}$, $L_n = 0.18 \mu\text{m}$, while each PMOS uses $W_p = 1.8 \mu\text{m}$, $L_p = 0.18 \mu\text{m}$.

The diffusion extension along the channel is again taken as

$$X_d = 3L_{\min} = 3 \times 0.18 \mu\text{m} = 0.54 \mu\text{m},$$

The corresponding source/drain areas and perimeters are

- **PMOS devices**

$$A_s = A_d = X_d W_p = 0.54 \times 1.8 = 0.972 \mu\text{m}^2,$$
$$P_s = P_d = 2X_d + 2W_p = 2 \times 0.54 + 2 \times 1.8 = 4.68 \mu\text{m}.$$

- **NMOS devices**

$$A_s = A_d = X_d W_n = 0.54 \times 0.9 = 0.486 \mu\text{m}^2,$$
$$P_s = P_d = 2X_d + 2W_n = 2 \times 0.54 + 2 \times 0.9 = 2.88 \mu\text{m}.$$

2.1.3 Sizing for the Sum (S) Transistors

For the Sum, all MOS devices share the same channel and diffusion lengths:

$$L = 0.18 \mu\text{m}, \quad X_d = 3L = 3 \times 0.18 = 0.54 \mu\text{m}.$$

The corresponding transistor widths are

$$W_n = 4 \times 0.9 = 3.6 \mu\text{m} \quad (\text{NMOS devices labeled } 4n),$$

$$W_p = 3 \times 1.8 = 5.4 \mu\text{m} \quad (\text{PMOS devices labeled } 3p).$$

From these values, the source/drain areas and perimeters of the Sum-network devices are

- **PMOS devices (3p)**

$$A_s = A_d = X_d W_p = 0.54 \times 5.4 = 2.916 \mu\text{m}^2,$$

$$P_s = P_d = 2X_d + 2W_p = 2 \times 0.54 + 2 \times 5.4 = 11.88 \mu\text{m}.$$

- **NMOS devices (4n)**

$$A_s = A_d = X_d W_n = 0.54 \times 3.6 = 1.944 \mu\text{m}^2,$$

$$P_s = P_d = 2X_d + 2W_n = 2 \times 0.54 + 2 \times 3.6 = 8.28 \mu\text{m}.$$

2.1.4 Sizing for the Carry-Out (C_{out}) Transistors

For the Carry-out path, the same channel and diffusion lengths are used:

$$L = 0.18 \mu\text{m}, \quad X_d = 0.54 \mu\text{m}.$$

The transistor widths are

$$W_n = 3 \times 0.9 = 2.7 \mu\text{m} \quad (\text{NMOS devices labeled } 3n),$$

$$W_p = 2 \times 1.8 = 3.6 \mu\text{m} \quad (\text{PMOS devices labeled } 2p).$$

The corresponding source/drain areas and perimeters are

- **NMOS devices (3n)**

$$A_s = A_d = X_d W_n = 0.54 \times 2.7 = 1.458 \mu\text{m}^2,$$

$$P_s = P_d = 2X_d + 2W_n = 2 \times 0.54 + 2 \times 2.7 = 6.48 \mu\text{m}.$$

- **PMOS devices (2p)**

$$A_s = A_d = X_d W_p = 0.54 \times 3.6 = 1.944 \mu\text{m}^2,$$

$$P_s = P_d = 2X_d + 2W_p = 2 \times 0.54 + 2 \times 3.6 = 8.28 \mu\text{m}.$$

Netlist for Full-Adder

Listing 9: LTspice netlist used for the full-adder

```
* Full Adder
.Include "D:\AUC\Uni\Fall 25\Microelectronics\Project\Saif and adham\CMOS 0.18\MOSFET_models_0p5_0p18-3.inc"

* Power Supply
VDD No1 0 DC 5V

*Inputs, testing DC
*Va   A    0   DC 5
*Vb   B    0   DC 0
*Vcin Cin  0   DC 5
```

* Input pulses

Va A 0 pulse(0 5 0 5n 5n 40n 100n)

Vb B 0 pulse(0 5 0 5n 5n 20n 50n)

Vcin Cin 0 pulse(0 5 0 5n 5n 10n 20n)

* Inverters

* InvA

Mpmos1 NotA A No1 No1 PMOSOP18 L=0.18u W=0.9u AD=0.486p AS=0.486p PD=2.88u PS=2.88u

Mnmos1 NotA A 0 0 NMOSOP18 L=0.18u W=0.9u AD=0.486p AS=0.486p PD=2.88u PS=2.88u

* InvB

Mpmos2 NotB B No1 No1 PMOSOP18 L=0.18u W=0.9u AD=0.486p AS=0.486p PD=2.88u PS=2.88u

Mnmos2 NotB B 0 0 NMOSOP18 L=0.18u W=0.9u AD=0.486p AS=0.486p PD=2.88u PS=2.88u

* InvCin

Mpmos3 NotCin Cin No1 No1 PMOSOP18 L=0.18u W=0.9u AD=0.486p AS=0.486p PD=2.88u PS=2.88u

Mnmos3 NotCin Cin 0 0 NMOSOP18 L=0.18u W=0.9u AD=0.486p AS=0.486p PD=2.88u PS=2.88u

* PUN - S

* A & notB & Cin

Mpmos4 mo4 A No1 No1 PMOSOP18 L=0.18u W=5.4u AD=2.916p AS=2.916p PD=11.88u PS=11.88u

Mpmos5 mo5 NotB mo4 mo4 PMOSOP18 L=0.18u W=5.4u AD=2.916p AS=2.916p PD=11.88u PS=11.88u

Mpmos6 S Cin mo5 mo5 PMOSOP18 L=0.18u W=5.4u AD=2.916p AS=2.916p PD=11.88u PS=11.88u

*notA & B & Cin

Mpmos7 mo7 NotA No1 No1 PMOSOP18 L=0.18u W=5.4u AD=2.916p AS=2.916p PD=11.88u PS=11.88u

Mpmos8 mo8 B mo7 mo7 PMOSOP18 L=0.18u W=5.4u AD=2.916p AS=2.916p PD=11.88u PS=11.88u

Mpmos9 S Cin mo8 mo8 PMOSOP18 L=0.18u W=5.4u AD=2.916p AS=2.916p PD=11.88u PS=11.88u

* A & B & notCin

Mpmos10 mo10 A No1 No1 PMOSOP18 L=0.18u W=5.4u AD=2.916p AS=2.916p PD=11.88u PS=11.88u

Mpmos11 mo11 B mo10 mo10 PMOSOP18 L=0.18u W=5.4u AD=2.916p AS=2.916p PD=11.88u PS=11.88u

Mpmos12 S NotCin mo11 mo11 PMOSOP18 L=0.18u W=5.4u AD=2.916p AS=2.916p PD=11.88u PS=11.88u

* notA & notB & notCin

Mpmos13 mo13 NotA No1 No1 PMOSOP18 L=0.18u W=5.4u AD=2.916p AS=2.916p PD=11.88u PS=11.88u

Mpmos14 mo14 NotB mo13 mo13 PMOSOP18 L=0.18u W=5.4u AD=2.916p AS=2.916p PD=11.88u PS=11.88u

Mpmos15 S NotCin mo14 mo14 PMOSOP18 L=0.18u W=5.4u AD=2.916p AS=2.916p PD=11.88u PS=11.88u

* PDN - S

Mnmos4 S A N1 N1 NMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mnmos5 S NotB N1 N1 NMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mnmos6 S Cin N1 N1 NMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mnmos7 N1 NotA N2 N2 NMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mnmos8 N1 B N2 N2 NMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mnmos9 N1 Cin N2 N2 NMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mnmos10 N2 A N3 N3 NMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mnmos11 N2 B N3 N3 NMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mnmos12 N2 NotCin N3 N3 NMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mnmos13 N3 NotA 0 0 NMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mnmos14 N3 NotB 0 0 NMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mnmos15 N3 NotCin 0 0 NMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u

* PUN - Cout

Mpmos16 mo16 NotA No1 No1 PMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mpmos17 Cout NotB mo16 mo16 PMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mpmos18 mo18 NotCin No1 No1 PMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mpmos19 Cout NotA mo18 mo18 PMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mpmos20 mo20 B No1 No1 PMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u
Mpmos21 Cout NotCin mo20 mo20 PMOSOP18 L=0.18u W=3.6u AD=1.944p AS=1.944p PD=8.28u PS=8.28u

* PDN - Cout

Mnmos16 Cout NotA N4 N4 NMOSOP18 L=0.18u W=2.7u AD=1.458p AS=1.458p PD=6.48u PS=6.48u
Mnmos17 Cout NotB N4 N4 NMOSOP18 L=0.18u W=2.7u AD=1.458p AS=1.458p PD=6.48u PS=6.48u
Mnmos18 N4 NotA N5 N5 NMOSOP18 L=0.18u W=2.7u AD=1.458p AS=1.458p PD=6.48u PS=6.48u
Mnmos19 N4 NotCin N5 N5 NMOSOP18 L=0.18u W=2.7u AD=1.458p AS=1.458p PD=6.48u PS=6.48u
Mnmos20 N5 NotB 0 0 NMOSOP18 L=0.18u W=2.7u AD=1.458p AS=1.458p PD=6.48u PS=6.48u
Mnmos21 N5 NotCin 0 0 NMOSOP18 L=0.18u W=2.7u AD=1.458p AS=1.458p PD=6.48u PS=6.48u

* Transient Analysis

.Tran 0.1n 100n
.Print Tran V(A) V(B) V(Cin) V(S) V(Cout)

* Timing measurements

.MEASURE Tran TpLH_Cout TRIG V(Cin) val=2.5 rise=2 targ V(Cout) val=2.5 rise=1
.MEASURE Tran TpHL_Cout TRIG V(Cin) val=2.5 fall=2 targ V(Cout) val=2.5 fall=1
.MEASURE Tran TpLH_Sum TRIG V(A) val=2.5 rise=1 targ V(S) val=2.5 rise=1
.MEASURE Tran TpHL_Sum TRIG V(A) val=2.5 fall=1 targ V(S) val=2.5 fall=1
.MEASURE Tran Tpd_Avg_Sum PARAM='(TpLH_Sum+TpHL_Sum)/2'

```

.MEASURE Tran Tpd_Avg_Cout PARAM='(TpLH_Cout+TpHL_Cout)/2'

* Power measurements
.MEASURE Tran PowerAvg avg I(VDD)*V(No1) from 0 to 100n
.MEASURE Tran Energy INTEG I(VDD)*V(No1) from 0 to 100n
.MEASURE Tran s_000 FIND V(s) at=80n
.MEASURE Tran s_011 FIND V(s) at=68n
.MEASURE Tran s_010 FIND V(s) at=60n
.MEASURE Tran s_001 FIND V(s) at=50n
.MEASURE Tran s_101 FIND V(s) at=45n
.MEASURE Tran s_100 FIND V(s) at=40n
.MEASURE Tran s_110 FIND V(s) at=20n
.MEASURE Tran s_111 FIND V(s) at=6n

.MEASURE Tran cout_000 FIND V(Cout) at=80n
.MEASURE Tran cout_011 FIND V(Cout) at=68n
.MEASURE Tran cout_010 FIND V(Cout) at=60n
.MEASURE Tran cout_001 FIND V(Cout) at=50n
.MEASURE Tran cout_101 FIND V(Cout) at=45n
.MEASURE Tran cout_100 FIND V(Cout) at=40n
.MEASURE Tran cout_110 FIND V(Cout) at=20n
.MEASURE Tran cout_111 FIND V(Cout) at=6n

.End

```

2.2 Functional Verification

2.2.1 Theoretical Test Cases

To confirm that the CMOS full adder behaves as intended, all eight input combinations of A , B , and C_{in} were considered. Table 1 lists the ideal (theoretical) values of the Sum output S and the Carry-out C_{out} for each case.

Table 1: Full-adder truth table

A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

2.2.2 Testing using DC

Listing 10: LTspice netlist snippet for DC test of the full adder

```
* Inputs, testing DC
Va    A    0    DC 5
Vb    B    0    DC 0
Vcin  Cin  0    DC 5
```

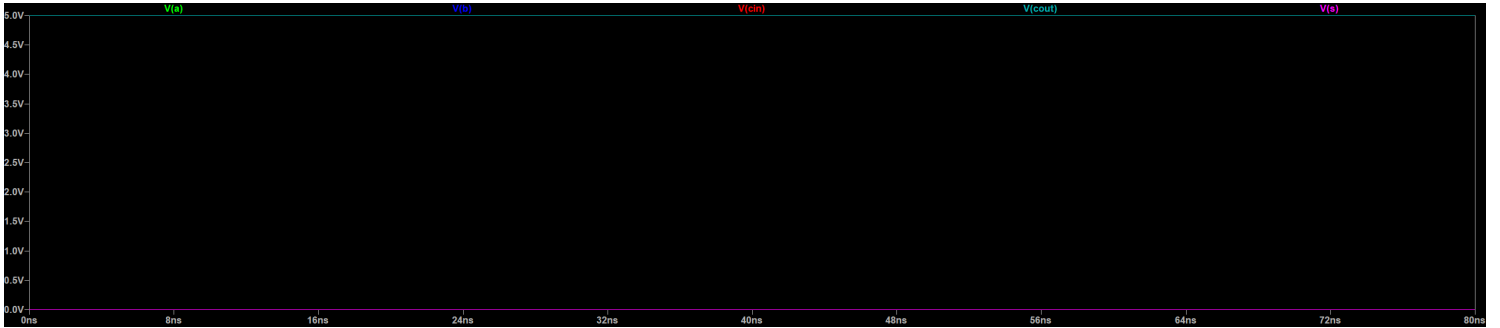


Figure 20: Full-adder DC testing Output

Guide:

- $V(s)$: pink
- $V(C_{out})$: cyan

We did this process for all cases and got the following output:

Table 2: Full-adder DC testing Output

A (V)	B (V)	C_{in} (V)	S (V)	C_{out} (V)
0	0	0	0	0
0	0	5	5	0
0	5	0	5	0
0	5	5	0	5
5	0	0	5	0
5	0	5	0	5
5	5	0	0	5
5	5	5	5	5

2.2.3 Testing using Pulse signals

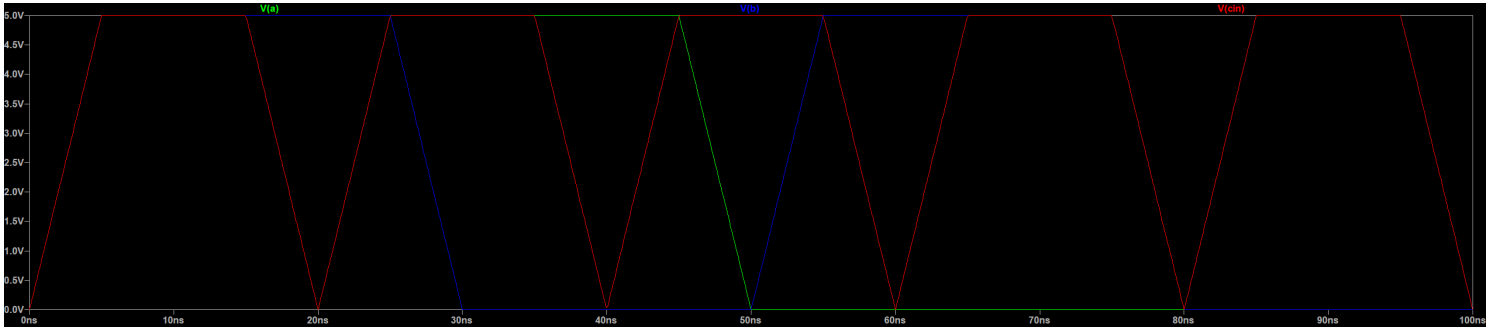


Figure 21: The three inputs A , B , C_{in}

Guide:

- $V(A)$: green
- $V(B)$: blue
- $V(C_{in})$: red

A	B	C	Time (n)
1	1	1	6
1	1	0	20
1	0	0	40
1	0	1	45
0	0	1	50
0	1	0	60
0	1	1	68
0	0	0	80

Table 3: Signal measurement times for full adder cases

2.2.4 Simulation Results

Transient simulations were then performed using the LTspice netlists for the Sum and Carry-out networks. For each input pattern, the steady-state voltages at S and C_{out} were recorded and translated into logic values: voltages close to V_{DD} are interpreted as logic high (1), while voltages near 0 V are taken as logic low (0).

```
tplh_cout=-2.06322132225e-08 FROM 2.25000000299e-08 TO 1.86778680741e-09
tphl_cout=7.97499289489e-10 FROM 3.75e-08 TO 3.82974992895e-08
tplh_sum=-6.94100656959e-10 FROM 2.50000011794e-09 TO 1.80589946098e-09
tphl_sum=-2.92243438633e-08 FROM 4.75000000751e-08 TO 1.82756562118e-08
tpd_avg_sum: '(TpLH_Sum+TpHL_Sum)/2'=-1.49592222601e-08
tpd_avg_cout: '(TpLH_Cout+TpHL_Cout)/2'=-9.9173569665e-09
poweravg: AVG(I(VDD)*V(No1))=-0.0209262277764 FROM 0 TO 1e-07
energy: INTEG(I(VDD)*V(No1))=-2.09262277764e-09 FROM 0 TO 1e-07
s_000: V(s)=0.000377346674213 at 8e-08
s_011: V(s)=-1.77854697904e-05 at 6.8e-08
s_010: V(s)=4.99463701248 at 6e-08
s_001: V(s)=4.99751472473 at 5e-08
s_101: V(s)=0.00048935329692 at 4.5e-08
s_100: V(s)=4.99741916124 at 4e-08
s_110: V(s)=0.000983535661362 at 2e-08
s_111: V(s)=5.0000009346 at 6e-09
cout_000: V(Cout)=0.00010291433864 at 8e-08
cout_011: V(Cout)=-0.0359822019241 at 6.8e-08
cout_010: V(Cout)=-3.02651806123e-05 at 6e-08
cout_001: V(Cout)=1.14462924004 at 5e-08
cout_101: V(Cout)=5.00002080869 at 4.5e-08
cout_100: V(Cout)=-3.21122189127e-05 at 4e-08
cout_110: V(Cout)=4.99981021881 at 2e-08
cout_111: V(Cout)=5.00009714127 at 6e-09
```

Figure 22: SPICE Output Log

The numerical results can be summarized as in Table 4.

Sum output S						Carry output C_{out}					
A	B	C_{in}	$S_{\text{(theory)}}$	$S_{\text{(applied)}}$	Time (ns)	A	B	C_{in}	$C_{\text{out (theory)}}$	$C_{\text{out (applied)}}$	Time (ns)
0	0	0	0	0.00038	80	0	0	0	0	0.00010	80
0	0	1	1	4.99751	50	0	0	1	0	1.14463	50
0	1	0	1	4.99464	60	0	1	0	0	-0.00003	60
0	1	1	0	-0.00002	68	0	1	1	1	-0.03598	68
1	0	0	1	4.99742	40	1	0	0	0	-0.00003	40
1	0	1	0	0.00049	45	1	0	1	1	5.00002	45
1	1	0	0	0.00098	20	1	1	0	1	4.99981	20
1	1	1	1	5.00000	6	1	1	1	1	5.00010	6

Table 4: Simulated voltages and measurement times for the Sum output S and Carry output C_{out} compared with theoretical logic values. Time is given in nanoseconds.

2.3 Simulation Results: Using SPICE Pulse source

The transient simulations were used to verify that the CMOS full adder behaves correctly for all input combinations of A , B , and C_{in} . The three inputs are applied as periodic pulse sources with different periods so that, over the simulation window, the circuit cycles through all eight possible input patterns.

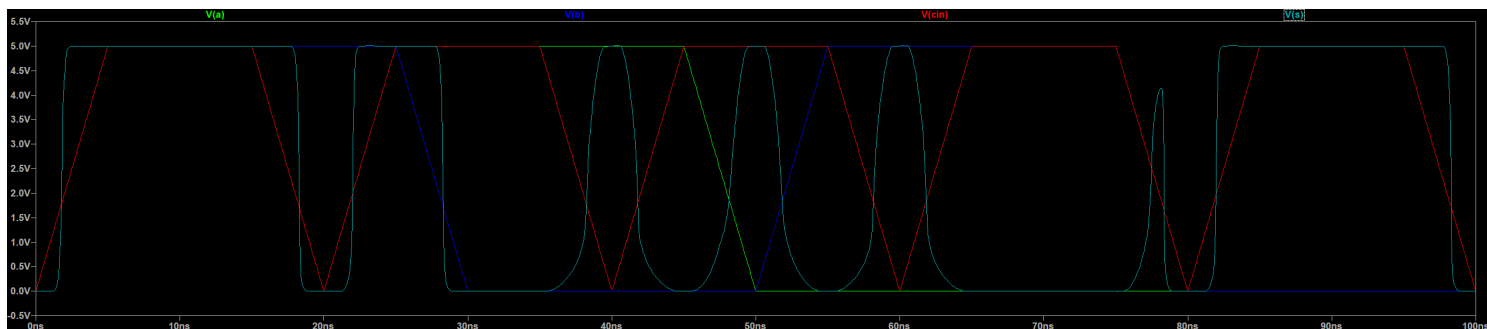


Figure 23: Transient response of the full adder showing the three inputs A , B , C_{in} and the sum output S .

Guide:

- $V(A)$: green
- $V(B)$: blue
- $V(C_{in})$: red
- $V(S)$: cyan

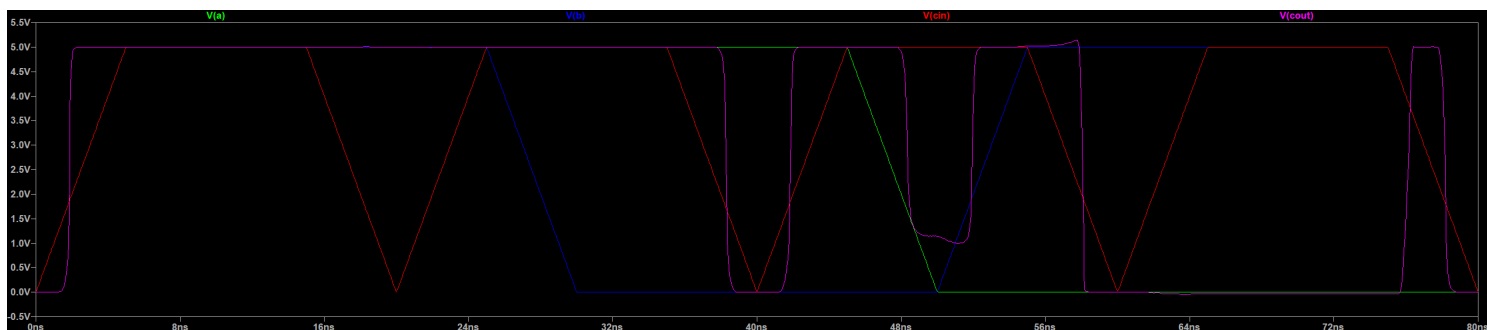


Figure 24: Transient response of the full adder showing the three inputs A , B , C_{in} and the carry output C_{out} .

Guide:

- $V(A)$: green
- $V(B)$: blue
- $V(C_{in})$: red
- $V(C_{out})$: pink

From the waveforms in Figures 24 and 23 we can make the following observations:

- The sum output S also follows the expected behaviour, switching between logic 0 and logic 1 whenever the parity (odd/even number of ones) of the inputs changes.

- As A , B , and C_{in} step through all eight input combinations, C_{out} toggles exactly according to the full-adder truth table: it is high whenever at least two of the three inputs are high.
- The simulated high and low levels of both S and C_{out} are very close to the supply rails, which indicates good noise margins for both outputs.
- A small glitch can be seen in S in the time window from 76 ns to 79 ns where B changes to zero. This is due to internal node capacitances or instability in the fall of B , causing a small inaccuracy in the sum signal output.
- Apart from this short glitch, the timing and logic levels of S and C_{out} match the theoretical behaviour, confirming that the designed full adder implements the intended Boolean functions.

2.4 Propagation Delay and Power Evaluation

The dynamic behaviour of the full adder was characterised in LTSPICE using `.measure` statements. For each output node, two delays are reported:

t_{pHL} : input makes a low-to-high transition, output falls,
 t_{pLH} : input makes a high-to-low transition, output rises.

Both are measured between the times when the input and output waveforms cross $V_{\text{DD}}/2$. The average propagation delay is

$$t_{\text{pd}} = \frac{t_{\text{pHL}} + t_{\text{pLH}}}{2}.$$

Sum output S

- High-to-low delay:

$$t_{\text{pHL},S} = -29.22434 \text{ ns} \quad (47.50000 \text{ ns} \rightarrow 18.27566 \text{ ns})$$

- Low-to-high delay:

$$t_{\text{pLH},S} = -0.69410 \text{ ns} \quad (2.50000 \text{ ns} \rightarrow 1.80590 \text{ ns})$$

- Average propagation delay:

$$t_{\text{pd},S} = \frac{t_{\text{pHL},S} + t_{\text{pLH},S}}{2} \approx -14.95922 \text{ ns}.$$

Carry output C_{out}

- High-to-low delay:

$$t_{\text{pHL},C_{\text{out}}} = 0.79750 \text{ ns} \quad (37.50000 \text{ ns} \rightarrow 38.29750 \text{ ns})$$

- Low-to-high delay:

$$t_{\text{pLH},C_{\text{out}}} = -20.63221 \text{ ns} \quad (22.50000 \text{ ns} \rightarrow 1.86779 \text{ ns})$$

- Average propagation delay:

$$t_{\text{pd},C_{\text{out}}} = \frac{t_{\text{pHL},C_{\text{out}}} + t_{\text{pLH},C_{\text{out}}}}{2} \approx -9.91736 \text{ ns}.$$

Power and energy

- Average power dissipation over the interval 0 to 1×10^{-7} s:

$$P_{\text{avg}} = -20.92623 \text{ mW} \quad (0 \text{ ns} \rightarrow 100 \text{ ns})$$

- Total energy drawn from the supply over the same interval:

$$E_{\text{tot}} = -2.09262 \text{ nJ} \quad (0 \text{ ns} \rightarrow 100 \text{ ns})$$

Remark. In the raw `.measure` log some delays appear with a negative sign, because the reported target time is numerically smaller than the trigger time. The data show that the Sum node S experiences a larger effective delay than C_{out} .

Contributions

- ⇒ **Adham Ali** – Led Part A of the project, including the basic CMOS inverter design, initial transistor sizing, preparation of figures, LTspice simulations **Part A**, and writing the report. Helped in **Part B**.
- ⇒ **Saif Abd Elfattah** – Led Part B, including the full-adder implementation, Sum and Carry network sizing, LTspice simulations **Part B**, and verification tables. Helped in **Part A**.